

Cooperative In-Vehicle Intelligent Voice Agents in Level 4 Automation

Cansu Demir

Department of Artificial Intelligence and Human Interfaces
University of Salzburg
Salzburg, Austria
cansu.demir@plus.ac.at

Abstract

Supervised by **Univ.-Prof. Dr. Alexander Meschtscherjakov** and **Univ.-Prof. Dr. Manfred Tscheligi** at the University of Salzburg, Austria.

Level 4 automated vehicles (AVs) are entering real-world urban traffic, where they must coexist and coordinate with human-driven vehicles. Although prior research has addressed automated driving scenarios, it largely focuses on isolated human-vehicle interaction or single-exposure studies, offering limited insight into cooperative behavior in mixed-traffic and multi-driver settings. This dissertation investigates voice-based interaction as a mechanism for communicating AV intent during cooperative driving situations. Cooperation is framed as a social and temporal process shaped by how AV passengers interpret voice-based communication and how this understanding manifests in vehicle behavior that surrounding human drivers must anticipate. Using a user-centered, mixed-methods approach, the research combines multi-driver driving simulation with a longitudinal simulator study. The work examines perceived safety, interpretation of cooperative intent, and coordination in mixed traffic, as well as how these processes evolve over repeated interaction. The dissertation contributes empirical evidence and methodological guidance for cooperative interaction in multi-driver simulation.

CCS Concepts

• **Human-centered computing** → **Laboratory experiments; User studies; Collaborative interaction.**

Keywords

In-Vehicle Intelligent Agents, Automated Driving, Human-Vehicle Interaction, SAE Level 4 Automation

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1 Introduction and Motivation

Advances in vehicle automation are fundamentally changing how humans interact with driving systems. In SAE Level 4 automation, vehicles can perform the complete driving task within defined operational domains, and human occupants are no longer expected to act as continuous drivers or safety fallbacks. While this reduces manual driving demands, it introduces new interaction challenges related to understanding, coordination, and cooperation between humans and automated vehicles (AVs).

Prior research on automated driving has highlighted issues such as reduced situational awareness, automation bias, and miscalibrated trust [5–7]. However, much of this work focuses on isolated human-vehicle interaction or single-driver scenarios. This perspective is insufficient for cooperative driving situations, such as merging, yielding, or negotiating priority, where AV behavior must be interpreted and anticipated by multiple human drivers in mixed traffic.

SAE Level 4 vehicles are already being deployed in real-world urban environments through commercial robotaxi services, for example in cities such as San Francisco. These systems operate in mixed traffic alongside human-driven vehicles rather than in fully automated environments. As the transition toward fully automated traffic will likely be gradual and uneven, cooperation between AVs and human drivers is not a temporary edge case but a persistent interaction challenge. Understanding how human drivers perceive, interpret, and adapt to AV behavior in such contexts is therefore critical for safe and acceptable automated mobility.

This dissertation investigates cooperative human-automation interaction in SAE Level 4 driving, with a specific focus on voice-based interaction as a means of communicating AV intent to AV passengers [6, 12]. Rather than treating drivers as passive recipients of automation, cooperation is conceptualized as an interactional process in which AV passengers and surrounding human drivers actively make sense of AV behavior during dynamic traffic situations.

Voice-based interaction is particularly relevant in SAE Level 4 contexts, where AV passengers may not maintain sustained visual attention to the road or in-vehicle displays. Spoken communication allows the automated system to convey intent or upcoming actions in a timely and situated manner, supporting passenger understanding without relying on visual interfaces. In this work, voice is treated not as a general-purpose conversational assistant, but as a designed communication channel for short, context-specific utterances during cooperative maneuvers.

Building on concepts from cooperative automation and human-machine interaction, such as mutual predictability, shared situation

awareness, and calibrated trust [16], this research examines how voice-mediated sensemaking inside the AV shapes passengers' understanding of cooperative intent, and how this, in turn, influences interaction and coordination with surrounding human drivers over repeated encounters.

2 Theoretical Grounding and Related Work

The transition toward highly AVs is reshaping human-machine interaction (HMI) in the automotive domain. As vehicles increasingly assume driving functions, human involvement shifts from active control to supervision and collaboration, requiring a move beyond command-based interaction toward cooperative models of interaction [16].

Research on cooperative automation provides a useful lens for understanding this transition, framing humans and machines as teammates who must coordinate their actions and maintain mutual understanding [1]. Foundational models such as Sheridan's levels of automation [15] and Endsley's situation awareness framework [8] highlight risks associated with reduced human involvement, including miscalibrated trust, automation bias, and disengagement. These concerns have been discussed extensively for SAE Levels 3-5 [9, 14, 17] and remain relevant in SAE Level 4 contexts, where drivers are no longer responsible for control but still need to understand and anticipate AV behavior in traffic.

Much existing research on automated driving interaction focuses on isolated human-vehicle interaction or single-driver scenarios. However, real-world deployments of SAE Level 4 systems increasingly take place in mixed-traffic environments, where AVs interact with multiple human drivers. In such settings, cooperation depends not only on technical vehicle behavior, but also on how that behavior is perceived, interpreted, and anticipated by surrounding road users. Despite its practical relevance, empirical research on cooperative interaction in multi-driver contexts remains limited.

Trust and perceived safety are central constructs in automated driving research [3–5, 11], yet they are often treated as static outcomes rather than as processes that evolve through repeated interaction. Many studies rely on single-exposure evaluations, offering limited insight into how drivers adapt to AV behavior over time. This limitation is particularly relevant for interaction modalities such as voice, where understanding, expectations, and reliance may develop gradually through experience.

In this context, in-vehicle intelligent agents (IVIs) have been proposed as mediators of interaction between humans and AVs. Prior work suggests that IVIs can support trust and user satisfaction when they are transparent and responsive [2, 10, 13]. However, existing systems typically address narrow functions—such as alerts or animations—and are rarely examined in cooperative, multi-driver, or longitudinal settings. Moreover, voice-based interaction is often treated as a generic interface component rather than as a situated channel for supporting passenger sensemaking during cooperative driving.

This dissertation builds on these strands of work by focusing on voice-based interaction as a mechanism for supporting cooperation with SAE Level 4 AVs in mixed traffic. By examining cooperative driving situations in multi-driver simulation and studying adaptation over time, the research addresses open questions in automotive

HCI concerning how internal, passenger-facing interaction influences cooperative behavior with surrounding human drivers.

3 Goals and Objectives

The overarching goal of this thesis is to understand how voice-based interaction can support cooperative interaction between SAE Level 4 AVs and human drivers in mixed-traffic environments. The research focuses on how voice-based communication addressed to AV passengers can shape cooperative behavior in scenarios where AVs and human-driven vehicles coexist for the foreseeable future.

Rather than developing fully autonomous agent systems, this work investigates voice-based IVIs as interaction concepts for communicating cooperative intent from within the AV. Cooperation is treated as a human-centered, social, and temporal process that unfolds through repeated interaction and adaptation over time.

To address this goal, the dissertation is structured around two complementary empirical studies: (1) a multi-driver driving simulation study examining cooperative interaction in mixed traffic, and (2) a longitudinal simulator study investigating adaptation to voice-based interaction by AV passengers across repeated encounters.

3.1 Objective 1: Perceived safety in cooperative multi-driver scenarios

The first objective is to examine how voice-based communication provided to AV passengers influences perceived safety during cooperative driving maneuvers, such as merging or yielding, in mixed-traffic situations. Using a multi-driver driving simulation, this objective focuses on subjective safety assessments, comfort, and risk perception among both AV passengers and surrounding manual drivers.

RQ1: How does voice-based communication to AV passengers influence perceived safety during cooperative driving maneuvers for both AV passengers and manual drivers?

3.2 Objective 2: Interpretation and coordination in cooperative driving

Building on the same multi-driver simulation study, the second objective investigates how AV passengers interpret voice-based communication and how this interpretation shapes coordination behavior in cooperative driving situations. This includes examining anticipation of AV behavior, timing of responses, and coordination dynamics between AVs and surrounding human-driven vehicles.

RQ2: How do AV passengers interpret voice-based communication of AV intent, and how does this interpretation influence coordination with surrounding manual drivers in cooperative driving scenarios?

3.3 Objective 3: Longitudinal adaptation to voice-based interaction

The third objective examines how AV passengers' perception, trust, and cooperative behavior evolve over repeated interactions with voice-based IVIs. Through a longitudinal simulator study, this objective focuses on learning effects, expectation formation, and changes in reliance or skepticism toward voice-based communication over time.

RQ3: How do AV passengers adapt their perception, trust, and cooperative behavior toward voice-based communication over repeated interactions?

4 Methodology

This research adopts a user-centered, mixed-methods approach to investigate voice-based IVIAs that support cooperative interaction in mixed-traffic environments involving SAE Level 4 AVs. Rather than developing full autonomous systems, the methodology focuses on designing and evaluating voice-based interaction concepts that communicate AV intent to AV passengers.

Cooperation is conceptualized as a social and temporal phenomenon, shaped by how human drivers and AV passengers perceive, interpret, and adapt to AV behavior. To capture both experiential and behavioral aspects of cooperation, the methodology combines multi-driver driving simulation, longitudinal evaluation, and simulator telemetry analysis, integrating quantitative and qualitative data.

4.1 Multi-driver simulation study

The primary empirical component of this thesis is a multi-driver driving simulator study designed to examine cooperative interaction in mixed traffic involving SAE Level 4 AVs and human-driven vehicles. The study is planned to involve approximately three groups of 15 participants driving simultaneously within the same simulated urban environment. Within each group, a subset of participants (e.g., around five) will experience the scenario as passengers in an AV, while the remaining participants drive manually controlled vehicles.

This setup approximates realistic mixed-traffic conditions in which AVs, their passengers, and human drivers coexist and interact. Cooperative maneuvers such as merging, yielding, or negotiating priority form the core interaction contexts.

Voice-based IVIA prototypes communicate AV intent exclusively to AV passengers through structured, context-specific spoken messages. Manual drivers do not receive voice-based interaction; their experience of cooperation emerges indirectly through the observable behavior of the AV in traffic. This design enables the investigation of how internal interaction within the AV shapes external coordination with surrounding human drivers.

The study is designed to support multiple analytical perspectives on the same interaction context, allowing perceived safety, interpretation of cooperative intent, and coordination behavior to be examined without requiring separate experimental setups.

Planned data collection includes:

- Simulator telemetry data (e.g., trajectories, speeds, distances, maneuver timing);
- Behavioral measures (e.g., response timing and coordination patterns);
- Subjective measures (e.g., perceived safety, comfort, trust, understanding);
- Post-task interviews oring participants' interpretation of AV behavior and voice-based communication.

4.2 Longitudinal simulator study

To examine how cooperation evolves over time, a longitudinal driving simulator study will be conducted using an existing high-fidelity driving simulator laboratory at the University of Salzburg. In this study, participants interact as AV passengers with a voice-based IVIA across multiple sessions, enabling investigation of learning effects, expectation formation, and changes in trust and cooperative behavior.

Planned data collection includes repeated self-report measures, interaction logs, and simulator telemetry data, complemented by follow-up interviews to capture how participants' mental models of voice-based communication develop over time.

4.3 Participants and analysis

Participants are licensed drivers with basic familiarity with automated driving concepts but without expert-level technical knowledge, reflecting realistic mixed-traffic conditions. Quantitative and qualitative data will be analyzed using appropriate statistical and thematic analysis methods, enabling triangulation between telemetry data, observed behavior, and self-reported experience.

While core study structures and measures are defined, specific parameters (e.g., sample sizes, scenario configurations, or message designs) may be refined iteratively based on pilot studies and practical constraints. Where feasible, study materials and anonymized datasets will be shared via open repositories to support transparency and reproducibility.

5 Expected Contribution

This dissertation contributes to research on human-centered interaction with SAE Level 4 automated vehicles by providing empirical and methodological insights into voice-based communication for cooperative driving in mixed-traffic environments. Rather than proposing a comprehensive agent architecture, the work focuses on understanding how specific interaction concepts shape human perception, behavior, and cooperation around automated vehicles.

A first contribution is empirical evidence on cooperative interaction in multi-driver contexts. Through controlled driving simulator studies, the research examines how AV passengers' interpretation of voice-based communication relates to observable vehicle behavior, and how this behavior is perceived and anticipated by surrounding manual drivers during cooperative maneuvers. The analysis focuses on perceived safety, interpretation of cooperative intent, and coordination in mixed traffic. By studying cooperation as a mediated social interaction involving multiple human drivers, the dissertation addresses limitations of prior work that relies on single-driver assumptions.

A second contribution is methodological, demonstrating how multi-driver driving simulation can be used to study cooperative automated driving under controlled yet dynamic conditions. By combining simulator telemetry data, behavioral measures, and qualitative insights, the work provides practical guidance for investigating interaction phenomena that emerge between AVs and multiple human road users, beyond isolated or scripted scenarios.

The third contribution concerns longitudinal understanding of voice-based interaction from the perspective of AV passengers. By examining repeated interactions over time, the research provides

insight into how trust, perceived safety, and cooperative expectations toward automated vehicle behavior evolve, highlighting learning effects and expectation formation that are often overlooked in single-exposure studies.

6 Mentoring Needs and Open Questions

While this dissertation builds on prior experience in automotive HCI, several open questions would benefit from targeted mentoring within the CHI community.

A first mentoring need concerns the positioning of cooperative driving research for a broader HCI audience. I would appreciate feedback on how findings from multi-driver simulation studies can be framed in ways that resonate beyond automotive HCI, particularly for researchers working on human-AI interaction, cooperative systems, and safety-critical interaction.

A second area relates to the design and reporting of longitudinal studies. Although adaptation over time is widely discussed in HCI, practical guidance on scoping longitudinal simulator studies, selecting meaningful measures, and communicating change over repeated interaction remains limited. Mentoring on balancing rigor, feasibility, and clarity in longitudinal evaluation would be especially valuable.

I am also interested in feedback on the methodological strengths and limitations of multi-driver simulation as an HCI method. Input on how to transparently communicate trade-offs between realism, control, and complexity and how to position this method within CHI expectations would help strengthen the dissertation.

Overall, I see the Student Mentoring Program as an opportunity to reflect critically on these questions, learn from others' experiences, and ensure that the remaining phases of my PhD are well scoped, well grounded, and clearly communicated.

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